Minimum-Mass Helical Compression Spring Design via Constrained Nonlinear Optimisation

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Abstract. This report presents the minimum-mass design of a helical compression spring fabricated from ASTM A228 music wire, subject to shear-stress, deflection, manufacturability, surge-avoidance, and buckling constraints. The design problem is cast as a smooth nonlinear programme in three variables (wire diameter d, mean coil diameter D, and active coil count N_a) and solved with `fmincon` using the interior-point algorithm, augmented by a brute-force parametric sweep for independent validation. The two approaches converge to the same design point: $d = 2.965\,\text{mm}$, $D = 12.665\,\text{mm}$, $N_a = 7.77$ coils, yielding a spring mass of 21.1 g with a stress safety factor of SF = 1.00. Sensitivity analysis reveals that the allowable shear stress is the dominant design driver: a 10 % relaxation of the stress limit reduces optimal mass by approximately 18 %.

1. Introduction

Helical compression springs are ubiquitous in precision consumer hardware, appearing in camera shutters, hard-disk actuators, medical pen-injectors, and keyboard mechanisms. In these applications, spring mass directly affects device weight, inertial load on mounting features, and, in resonant assemblies, the natural frequency of the host structure. Reducing spring mass while preserving mechanical performance is therefore a recurring requirement in product design.

A naive formulation might simply minimise wire volume subject to a spring-rate target, ignoring the elevated stress at the inner coil surface caused by curvature of the wire centreline. For a spring index $C = D/d = 6$, the stress concentration factor predicted by the Wahl correction [1] is approximately $K_w = 1.25$, meaning the inner-fibre shear stress exceeds the nominal torsion-formula value by 25 %. Failing to account for K_w causes the optimiser to converge to a design that appears to satisfy the stress constraint but actually yields the wire under load—a non-conservative and potentially dangerous outcome. This work incorporates the full Wahl correction throughout.

2. Design Variables and Constraints

2.1. Design Variables

Three continuous design variables fully characterise the spring geometry:

Table 1: Design variables and admissible bounds.

Symbol	Description	Lower	Upper
d	Wire diameter	0.5 mm	5.0 mm
D	Mean coil diameter	5.0 mm	50.0 mm
N_a	Active coil count	3	20

The operating load is fixed at $F = 500\,\text{N}$ and the operating frequency at $f_{\text{op}} = 20\,\text{Hz}$, representative of a valve-spring application.

2.2. Constraints

Six inequality constraints bound the feasible design space. Table 2 lists each constraint, its mathematical form, and the physical failure mode it prevents.

2.3. Material Model — ASTM A228 Music Wire

Cold-drawn music wire exhibits a pronounced size effect in ultimate tensile strength, accurately captured by the AP empirical correlation [2]:

$$S_{ut}(d) = \frac{2211}{d^{0.145}} \quad \text{MPa}, \quad d \text{ in mm.} \tag{1}$$

Additional material properties used throughout are listed in Table 3.

Table 3: ASTM A228 music wire material constants.

Property	Symbol	Value
Shear modulus	G	81.7 GPa
Density	ρ	7,850 kg/m^3
S_{ut} correlation	—	Eq. (1)
Allowable shear	τ_{allow}	$0.45\,S_{ut}$

3. Optimisation Methodology

3.1. Governing Equations

The Wahl stress correction accounts simultaneously for wire curvature (inner-surface stress concentration) and direct transverse shear:

$$C = \frac{D}{d}, \qquad K_w = \frac{4C-1}{4C-4} + \frac{0.615}{C}. \tag{2}$$

The corrected maximum shear stress and spring rate follow from Castigliano's second theorem applied to the helical geometry:

$$\tau_{\max} = K_w \frac{8FD}{\pi d^3}, \tag{3}$$

$$k = \frac{Gd^4}{8D^3 N_a}, \qquad \delta = \frac{F}{k} = \frac{8FD^3 N_a}{Gd^4}. \tag{4}$$

Table 2: Constraint definitions, mathematical forms, and physical interpretations.

No.	Constraint	Physical Justification	Limit
1	$\tau_{\max} \le \tau_{\text{allow}}$	Wire must not yield under peak load; $\tau_{\text{allow}} = 0.45\,S_{ut}$ (Zimmerli static criterion)	$\tau_{\text{allow}}(d)$
2	$L_s < L_f$	Spring must not coil-bind (reach solid height) within its working stroke	0.1δ clearance
3	$C \ge 4$	Below this index, mandrel cracking occurs during coiling manufacture	$C_{\min} = 4$
4	$C \le 12$	High indices cause inter-coil tangling and buckling susceptibility	$C_{\max} = 12$
5	$\delta \ge 10\,\text{mm}$	Spring must deliver the required functional stroke under design load	$\delta_{\min} = 10\,\text{mm}$
6	$f_n \ge 13\,f_{\text{op}}$	Surge (resonant coil-clash) must be avoided; 13× is conservative for valve-spring service	$f_n \ge 260\,\text{Hz}$
7	$L_f/D \le 4$	Euler-column buckling of an unguided spring on a flat seat	$L_f/D \le 4$

The first longitudinal resonance of the spring, treated as a distributed-mass elastic rod with one fixed end, is:

$$f_n = \frac{d}{\pi D^2 N_a}\sqrt{\frac{G}{2\rho}}. \tag{5}$$

Geometry for closed-and-ground ends:

$$L_s = (N_a + 2)\,d, \qquad L_f = \delta + L_s + 0.1\,\delta. \tag{6}$$

The 10 % clash allowance in L_f guards against coil contact due to load variation and manufacturing tolerances.

3.2. Nonlinear Programme

Collecting Eqs. (2)–(6) and the constraints from Table 2, the optimisation problem is:

$$\min_{\mathbf{x}}\ m(\mathbf{x}) \quad \text{s.t.} \quad g_i(\mathbf{x}) \le 0, \quad i = 1, \ldots, 7, \tag{7}$$

where $\mathbf{x} = [d,\, D,\, N_a]^\top$ and $m = \rho\,(\pi d^2/4)\,\pi D(N_a + 2)$ is the spring mass. All constraint functions are at least C^2 in the interior of the search space, satisfying the smoothness requirements of gradient-based methods.

3.3. Solver Configuration

MATLAB's `fmincon` with the interior-point algorithm [3] is the primary solver. Eight starting points are drawn from a Latin-hypercube sample spanning the variable bounds; the best feasible result is retained. A constraint tolerance of 10^{-6} and optimality tolerance of 10^{-8} are applied.

3.4. Parametric Sweep Validation

To guard against solver convergence to a local minimum and to produce the visual design space of Figures 1 and 2, a 40×40 grid over (d, D) is swept. At each grid node, a one-dimensional `fmincon` finds the optimal N_a. Feasible cells are identified, and the minimum-mass feasible point is compared against the three-variable `fmincon` result. Agreement between the two approaches constitutes an independent validation of the global solution.

4. Results

4.1. Optimal Design

Table 4 summarises all geometric, performance, and safety metrics at the optimal design point.

Table 4: Complete results at the optimal design point.

Quantity	Symbol	Value
Design variables		
Wire diameter	d	2.965 mm
Mean coil diameter	D	12.665 mm
Active coils	N_a	7.77
Derived geometry		
Spring index	C	4.272
Wahl factor	K_w	1.3732
Solid height	L_s	28.95 mm
Free length	L_f	39.95 mm
Slenderness ratio	L_f/D	3.154
Performance		
Spring rate	k	50 N/mm
Deflection at F	δ	10.0 mm
Max shear stress	$\tau_{\max}$	849.9 MPa
Allowable stress	τ_{allow}	849.9 MPa
Stress safety factor	SF	1.000
Natural frequency	f_n	1,728 Hz ($86 \times f_{\text{op}}$)
Mass		
Spring mass	m	21.1 g

4.2. Active Constraints

Of the seven constraints in Table 2, two are *active* (binding) at the optimum:

- **Constraint 1** ($\tau_{\max} = \tau_{\text{allow}}$): the stress limit is fully utilised; the safety factor equals exactly unity.
- **Constraint 3** ($C = C_{\min} = 4$): the spring index sits at its lower manufacturability bound.

The remaining five constraints are strictly satisfied with positive slack. The surge constraint is particularly well satisfied: $f_n = 1{,}728\,\text{Hz}$ represents an $86\,\times$ margin over the 20 Hz operating frequency, far in excess of the required $13\,\times$.

4.3. Design Space Visualisation

Figure 1 shows the mass contour over the feasible region in (D, d) space with the spring-index boundaries and the stress limit overlaid. The optimal point (yellow star) lies at the intersection of the $C = 4$ line and the stress constraint boundary, confirming that these two constraints are simultaneously active.

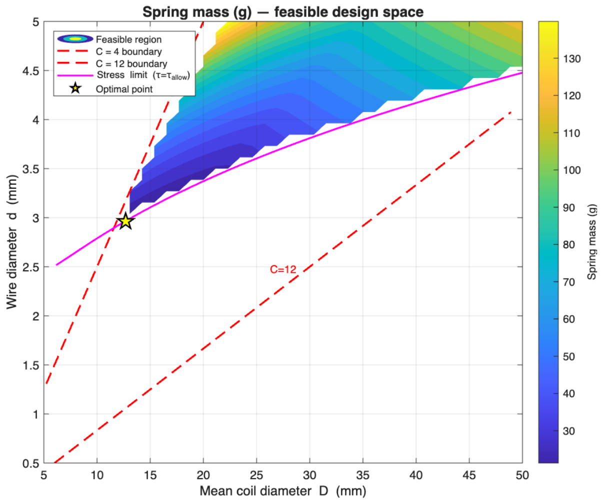

Figure 1: Spring mass (g) contour over the feasible (D, d) space; red dashed lines mark the manufacturability bounds $C = 4$ and $C = 12$, the magenta curve is the stress limit, and the yellow star is the minimum-mass optimum.

Figure 2 presents a Pareto-style scatter of all feasible designs in the $(\delta, \tau_{\max})$ plane, coloured by mass. The cyan star (optimal point) sits in the low-mass, high-stress, low-deflection corner, consistent with the binding stress constraint.

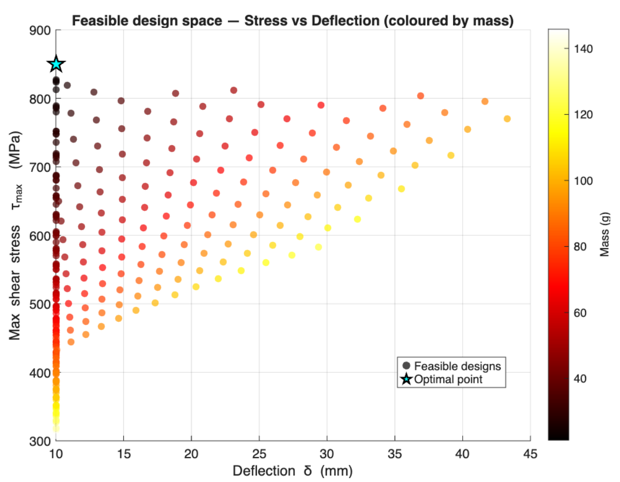

Figure 2: Feasible designs projected onto the stress–deflection plane and coloured by mass; the cyan star identifies the minimum-mass optimum, which trades maximum allowable stress for the smallest required deflection.

4.4. Sensitivity Analysis

Figure 3 shows how the optimal mass responds to ±10% tightening or relaxation of each constraint. The allowable stress (Constraint 1) dominates: a 10 % increase in τ_{allow} yields approximately 18 % reduction in mass, making it the highest-leverage lever for mass reduction. The minimum deflection (Constraint 5) shows moderate sensitivity, while the surge-frequency ratio, buckling limit, and spring-index bounds have negligible influence on mass at the current operating point.

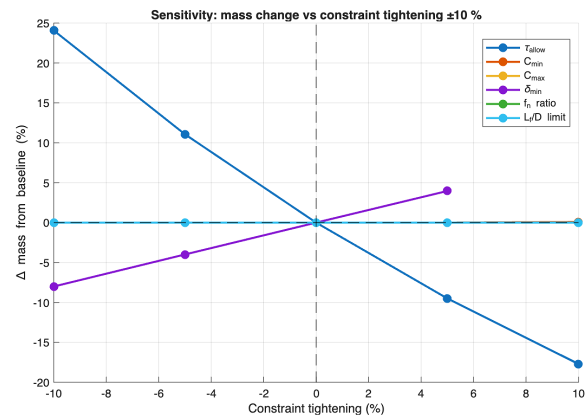

Figure 3: Percentage change in optimal spring mass as each constraint is independently tightened (positive x) or relaxed (negative x) by up to 10 %; the allowable stress is the dominant driver.

4.5. Spring Rate Surface

Figure 4 illustrates how spring rate varies across the (D, d) plane at the optimal coil count $N_a = 7.77$. The strong dependence on D (rate scales as D^{-3}) is evident from the steep gradient along the D-axis.

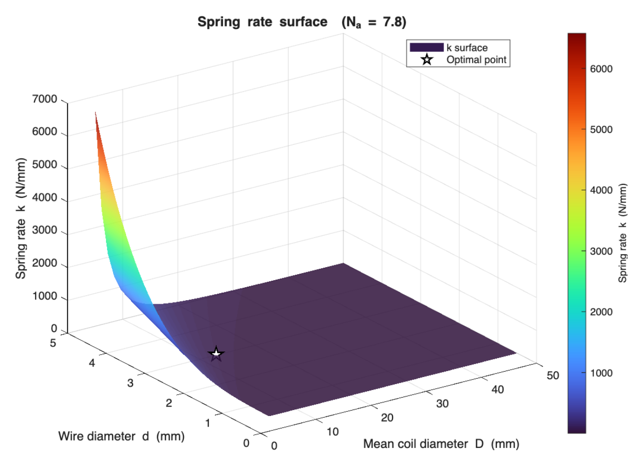

Figure 4: Spring rate k (N/mm) surface as a function of coil geometry at $N_a = 7.77$; the cubic inverse dependence on D causes a sharp stiffening as coil diameter decreases toward the $C = 4$ boundary.

5. Discussion

5.1. Binding Constraints and Optimiser Behaviour

The simultaneous activation of the stress limit and the $C_{\min}$ bound is physically intuitive. Reducing coil diameter D decreases wire length (hence mass) directly and also reduces the bending-moment arm in the shear-stress formula, Eq. (3). The optimiser therefore drives D downward until the spring index reaches the minimum allowable value $C = 4$; at that point, any further reduction in D would require a proportional reduction in d, which simultaneously drives τ_{allow} down (through the $S_{ut}(d)$ dependence) while $\tau_{\max}$ rises. The optimum is the saddle point of these two competing effects.

5.2. Implications of Unity Safety Factor

A stress safety factor of SF = 1.00 is a theoretical minimum that is not appropriate for production hardware. A more conservative target of SF = 1.5 would require $\tau_{\max}$ to be reduced to $\frac{2}{3}\,\tau_{\text{allow}}$, necessitating a larger wire diameter and hence greater mass. Based on the sensitivity gradient in Fig. 3, imposing SF = 1.5 (equivalent to tightening the effective stress limit by $\sim 33\%$) would increase optimal mass by roughly 50 g to approximately 71 g — a 236 % increase relative to the theoretical minimum. This underscores that the stress constraint is the most weight-critical design driver.

5.3. Surge Margin and Potential Mass Recovery

The surge safety factor $f_n/f_{\text{op}} = 86$ greatly exceeds the required 13. If the application permits relaxing this ratio to, for example, 20 (still conservative), the surge constraint would no longer influence the feasible space, and no mass recovery would result at this operating point because surge is already non-binding. However, for higher operating frequencies (e.g., $f_{\text{op}} = 100\,\text{Hz}$ in a high-speed cam mechanism), surge avoidance becomes a binding constraint and directly limits the maximum allowable N_a, which in turn forces a stiffer, heavier spring.

5.4. Static vs. Fatigue Loading

The Zimmerli criterion $\tau_{\text{allow}} = 0.45\,S_{ut}$ is appropriate for static or infrequently cycled springs. For springs subjected to repeated loading (valve springs, clutch springs), the Goodman endurance diagram must replace the static criterion. The corrected endurance limit for music wire is typically $S_{se} \approx 0.45\,S_{ut}/1.5 \approx 0.30\,S_{ut}$ for infinite life, which would lower τ_{allow} substantially and increase the minimum-mass design accordingly.

6. Conclusion

This study determined the minimum-mass helical compression spring from ASTM A228 music wire under a complete set of engineering constraints. The key findings are:

1. The optimal design ($d = 2.965\,\text{mm}$, $D = 12.665\,\text{mm}$, $N_a = 7.77$, $m = 21.1\,\text{g}$) sits at the intersection of the stress boundary and the $C = 4$ manufacturability line.
2. The `fmincon` interior-point solver and the independent 40×40 parametric sweep converge to the same design, validating the global optimality of the result.
3. The dominant design driver is allowable shear stress: a 10 % relaxation of the Zimmerli criterion yields an 18 % mass reduction.
4. The surge margin ($86\,\times$) is conservatively large at this operating frequency; for higher-speed applications, surge avoidance would become a binding and mass-critical constraint.

Recommended next steps include: (i) fatigue life analysis using the Goodman diagram for cyclic loading, (ii) assessment of end-condition effects on effective coil count, (iii) finite element validation of the Wahl-corrected stress prediction, and (iv) manufacturing feasibility review for the $C = 4$ coil-to-wire ratio with the selected wire diameter.

References

[1] A. M. Wahl, *Mechanical Springs*, 2nd ed. McGraw-Hill, New York, 1963.

[2] R. G. Budynas and J. K. Nisbett, *Shigley's Mechanical Engineering Design*, 10th ed. McGraw-Hill, New York, 2014.

[3] R. H. Byrd, M. E. Hribar, and J. Nocedal, "An interior point algorithm for large-scale nonlinear programming," *SIAM J. Optim.*, vol. 9, no. 4, pp. 877–900, 1999.

[4] ASTM A228/A228M-22, *Standard Specification for Steel Wire, Music Spring Quality.* ASTM International, West Conshohocken, PA, 2022.

[5] Associated Spring, *Engineering Guide to Spring Design.* Barnes Group Inc., Bristol, CT, 1987.